



LEIYANG HEXI ANIMAL MANURE COMPOSTING PROJECT



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1 PROJECT DETAILS

1.1 Summary Description of the Project

Leiyang Hexi Animal Manure Composting Project (hereafter referred to as “the Project”) is a centralized manure treatment plant established in Group 1, Wufeng Village, Xiaoshui Town, Leiyang City, Hengyang City, Hunan Province, People's Republic of China, and is developed by Leiyang Cailun Contemporary Agricultural Science and Technology Park Development Co., Ltd. (hereafter referred to as “the Project Proponent” or “the PP”).

The project recycles animal manure from local swine farms in Leiyang City and removes solids through a solid-liquid separation system. The separated solids are further used to produce organic solid fertilizers through aerobic composting, effectively eliminating the potential for the manure solids to produce methane emissions. The separated liquids are treated in water treatment systems to produce liquid fertilizers without being discharged into the fluvial system.

Prior to the implementation of the project, manure from swine barns was left to decay in uncovered lagoons for uncontrolled anaerobic decay and was used seasonally for irrigation. The lagoons were barely fully drained and cleaned, and the methane produced was not recovered and was emitted directly to the atmosphere.

The project activity, which introduces an animal manure composting system associated with solid separation techniques, avoids methane emissions through the separation of manure solids that are prevented from entering uncovered anaerobic lagoons, and further eliminates any potentials for methane emissions from the manure solids, as well as lowers the risk of exceeding nutrient application limits in the irrigation lands. The project is estimated to generate 52,500 tonnes of solids separated from swine manure and produce 30,000 tonnes of organic solid fertilizers per year.

The project implementation date was on 28-Mar-2022. It is estimated that during the first 7-year project crediting period (28-Mar-2022 to 27-Mar-2029), the total GHG emission reductions will reach to 419,587 tCO₂e, with annual emission reductions of 59,941 tCO₂e.

1.2 Sectoral Scope and Project Type

The project falls under Sectoral Scope 13: Waste handling and disposal.

The project is not an AFOLU project and is not a grouped project.

1.3 Project Eligibility

The scope of the VCS Program includes:

- 1) The seven Kyoto Protocol greenhouse gases: The Project is expected to avoid emissions of methane, which is a greenhouse gas designated by the Kyoto Protocol. Methane emissions from the anaerobic animal manure management system in the baseline scenario will be avoided or reduced in the project scenario.
- 2) Ozone-depleting substances: Not Applicable.
- 3) Project activities supported by a methodology approved under the VCS Program through the methodology approval process: Not Applicable.
- 4) Project activities supported by a methodology approved under a VCS approved GHG program, unless explicitly excluded under the terms of Verra approval: The applied methodology AMS-III.Y (Version 04.0) of the project is a methodology approved under CDM Program, which is a VCS approved GHG program.
- 5) Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements: Not Applicable.

The project does not belong to projects that can reasonably be assumed to have generated GHG emissions primarily for the purpose of their subsequent reduction, removal, or destruction.

Meanwhile, the project does not belong to the project activities excluded in Table 1 (Excluded Project Activities) of VCS Standard 4.5.

Therefore, the project is eligible under the scope of VCS program.

1.4 Project Design

The project is not a grouped project, and includes a single location only.

- ☒ The project includes a single location or installation only
- ☐ The project includes multiple locations or project activity instances, but is not being developed as a grouped project
- ☐ The project is a grouped project

Eligibility Criteria

Not applicable.

1.5 Project Proponent

Provide contact information for the project proponent(s). Copy and paste the table as needed.

Organization name	Leiyang Cailun Contemporary Agricultural Science and Technology Park Development Co., Ltd.
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Contact person	Jun Li
Title	Project Manager
Address	Group 4, Dajin Village, Caizichi Street, Leiyang City, Hengyang City, Hunan Province, People's Republic of China
Telephone	+86 2123019950
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1.6 Other Entities Involved in the Project

Organization name	Climate Bridge (Shanghai) Ltd.
Role in the project	Consultancy
Contact person	Zhiwen Gao
Title	General Manager
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1.7 Ownership

The project owner is Leiyang Cailun Contemporary Agricultural Science and Technology Park Development Co., Ltd., who has the legal right to control and operate the project activity. The business license, approval of the Environmental Impact Assessment (EIA) and the equipment purchasing contract are evidence of the ownership of the project and carbon credits generated.

1.8 Project Start Date

The start date of the Project is on 28-Mar-2022, coinciding with the date when the solid separation system and the composting system were simultaneously and officially put into operation.

1.9 Project Crediting Period

The project adopts a 7-year crediting period that is twice renewable for a total of 21 years. The first crediting period of the project is from 28-Mar-2022 to 27-Mar-2029 (both days inclusive).

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals of the project are:

- ☐ <20,000 tCO₂e/year
- ☒ 20,000 – 100,000 tCO₂e/year
- ☐ 100,001 – 1,000,000 tCO₂e/year
- ☐ >1,000,000 tCO₂e/year

Project Scale	
Project	X
Large project	

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
28-Mar-2022 - 31-Dec-2022	45,818
01-Jan-2023 - 31-Dec-2023	59,941
01-Jan-2024 - 31-Dec-2024	59,941
01-Jan-2025 - 31-Dec-2025	59,941
01-Jan-2026 - 31-Dec-2026	59,941
01-Jan-2027 - 31-Dec-2027	59,941
01-Jan-2028 - 04-Sep-2028	59,941
01-Jan-2029 - 27-Mar-2029	14,123
Total estimated ERs	419,587
Total number of crediting years	7
Average annual ERs	59,941

1.11 Description of the Project Activity

The project serves to reduce GHG emissions in local swine farms across Leiyang City through the following procedures.

- 1) Manure collection

When receiving requests for manure treatment from farmers in the municipal area of Leiyang City, the project owner sends vacuum trucks to the designated farms to collect fresh manure slurry. The manure slurry is transported airtightly to the solid separation system at the project site.

2) Solid separation system

The solid separation system is installed in the solid fertilizer workshop and is mainly comprised of two main components: a pre-treatment tank and a mechanical solid-liquid separator with screening and screw-pressing functions:

- Vacuum trucks pump the manure slurry to an enclosed pre-treatment tank with an effective volume of 1500m³, where auxiliary materials such as PH adjusters may be added to the manure slurry to meet the needs of solid separation. The pre-treated manure slurry usually remains a high moisture content (70% - 80%).
- The pre-treated manure slurry is then discharged onto the top of the screening segment of the solid-liquid separator, which utilizes vibrating inclined screens that allows dilute manure slurry to flow through the screen openings. The flowing-down manure slurry from the screening segment is then fed in at one end of the screw-pressing segment. The screw compresses the manure slurry to further force water out, the separated solids usually have a moisture content between 40% and 60%.

No flocculants are needed before or during separation procedures. The separated solids are aerobically composted on-site before any secondary application, while the separated liquids are diverted via pipes to the liquid fertilizer workshop, where a water management system that combines aerobic and anaerobic treatments is installed to produce organic liquid fertilizers.

3) Composting system

The separated solids coming off the solid separation system immediately enter the composting system in the same workshop without the need for storage. Auxiliary materials such as fermentation strains and agricultural waste-based ingredients (plant ash, straw, rich hull and etc.) may be added to the separated solids to meet the need of composting. The composting system consists of procedures including crushing, stirring and screening, all of which are carried out in an aerobic environment, effectively eliminating the potential for the manure solids to product methane emissions. The finished products are weighed, compressed and packed into baled bags by a baler in the solid fertilizer workshop and are available for sale. The annual productivity of solid fertilizers of the project reaches to 30,000 t/year.

The technical parameters of main equipment used in this project are shown in the table below:

Equipment	Type	Number
Solid-liquid Separator	LX-950K-3Q-20hDAF-20	2
Compost turner	RY-ZFP40B	1

1.12 Project Location

The Project is located in Group 1, Wufeng Village, Xiaoshui Town, Leiyang City, Hengyang City, Hunan Province, People's Republic of China, and the project site is located at north latitude of $26^{\circ}20'50.0''$ and east longitude of $112^{\circ}48'35.8''$ specifically. The geographic location of the project is depicted in Figure 1.

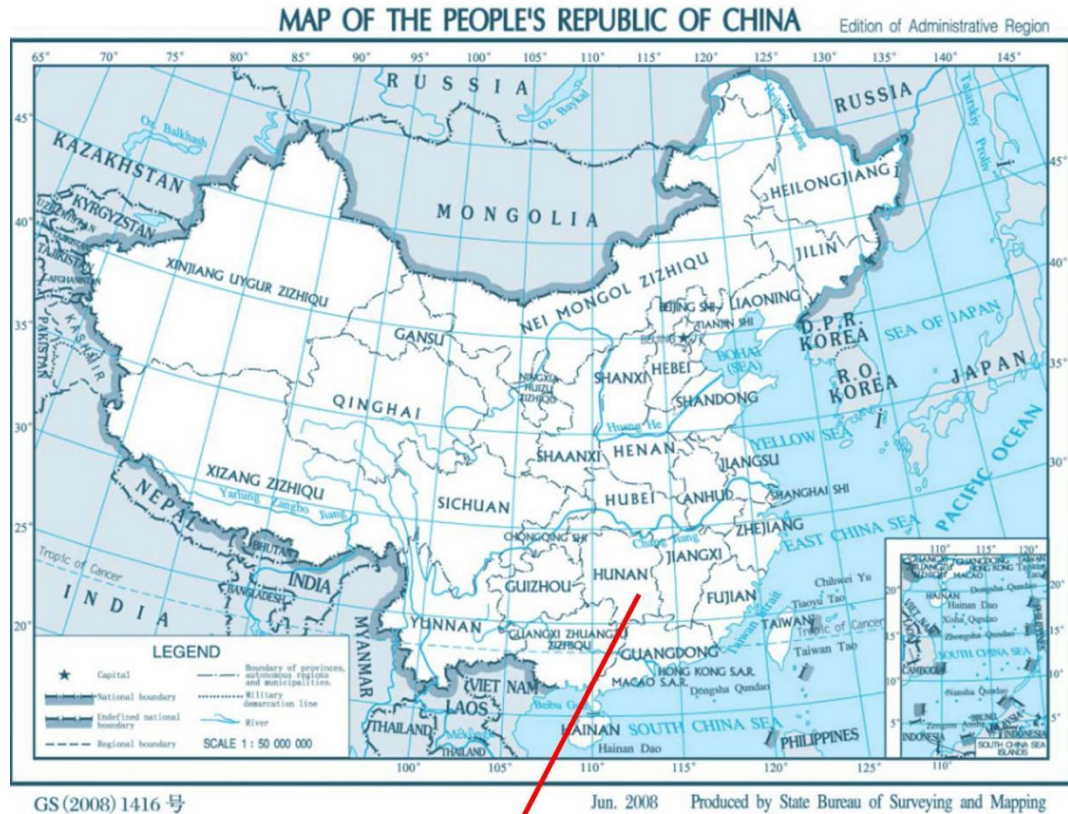


Figure 1 Location of the project

1.13 Conditions Prior to Project Initiation

Prior to the project initiation, animal manure at the local swine farms was left to decay in anaerobic manure management system (uncovered anaerobic lagoons) for a long term, and methane is emitted to the atmosphere directly without any methane recovery and destruction facility. The conditions prior to project implementation are the same as the baseline scenario, which would further be expanded upon in Section 3.4 of this PD.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project complies with following relevant laws and regulations in China:

- 1) Environmental Protection Law of People's Republic of China¹
- 2) Law of the People's Republic of China on the Prevention and Control of Solid Waste Pollution²
- 3) Regulations on prevention and control of pollution from large scale livestock and poultry breeding³
- 4) Catalogue for the Guidance of Industrial Structure Adjustment (2019 revision)⁴

The project has obtained the project approval and EIA approval from local government authorities: Agricultural Bureau of Hengyang City, and Ecology and Environment Bureau of Hengyang City. The two approvals well demonstrate that local government permits the construction of the project. Consequently, the project is compliance with laws, status and other regulatory frameworks.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project has neither been registered nor seeking registration under any other GHG programs. The project is seeking registration only under VCS program.

1.15.2 Projects Rejected by Other GHG Programs

The project has not been rejected by any GHG programs.

¹ http://www.china-npa.org/uploads/1/file/public/201803/20180327160313_j8uwkhdsgn.pdf

² https://www.mee.gov.cn/ywgz/fgbz/fl/202004/t20200430_777580.shtml

³ http://www.gov.cn/flfg/2013-11/26/content_2535095.htm

⁴ http://www.gov.cn/xinwen/2019-11/06/content_5449193.htm

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

Does the project reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading?

☐ Yes

☒ No

1.16.2 Other Forms of Environmental Credit

Has the project sought or received another form of GHG-related credit, including renewable energy certificates?

☐ Yes

☒ No

Supply Chain (Scope 3) Emissions

As per Clarification to VCS Program Rules and Requirements issued on 31-May-2023⁵, projects are not required to complete the sections in the affected VCS project templates that relate to Scope 3 emissions double claiming until the effective date of the revised requirements of 01-Jan-2024. Therefore, this section is not applicable.

1.17 Sustainable Development Contributions

The Project contributes to sustainable development in the following ways:



SDG 8 “Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all”⁶.

China continuously improves the quality and efficiency of development. In-depth implementation of the innovation-driven development strategy, the rapid development of small and medium-sized enterprises. Adhering to the policy of giving priority to employment, the unemployment rate has remained at a low level. By coordinating epidemic prevention and control and economic and social development, it has become the only major economy to achieve positive growth in 2020 and has made positive contributions to the recovery of the global economy.

The project will provide new job opportunities and increase tax revenue, which will have a positive effect on the local economy. This contributes to

⁵ <https://verra.org/wp-content/uploads/2023/06/VCS-Clarification-on-Scope-3-Emissions-Double-Claiming-Requirements-31-May-2023.pdf>

⁶ <https://sdgs.un.org/goals/goal8>

one of China's actions for promoting sustainable developing: "by 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value".



SDG 12 "Ensure sustainable consumption and production patterns"⁷.

China has made great efforts to develop a circular economy, publicizing, encouraging and promoting economical consumption patterns. To initially control the production and management of industrial solid waste and municipal waste. Through prevention, reduction, recycling and reuse, the generation of waste is greatly reduced, the recycling level of main waste is significantly improved, and the dependence on primary resources is reduced.

The project introduces the aerobic composting system that is designed to compost animal manure from local swine farms and to produce organic fertilizers. Thus, the project will achieve Target 12.5 of SDG 12 Ensure sustainable consumption and production patterns. This contributes to achieve one of China's stated sustainable development priorities "By 2030, substantially reduce waste generation through prevention, reduction, recycling and reuse".



SDG 13 "Take urgent action to combat climate change and its impacts"⁸.

In 2020, China's energy consumption per unit of GDP was reduced by 24.4% compared with 2012; carbon dioxide emissions per unit of GDP reduced by 18.8% compared with 2015 and 48.4% compared with 2005, all of which have already fulfilled China's commitment to the international community in 2020 ahead of schedule.

The project will reduce CH₄ emissions from animal manure through removing the volatile solids and composting the separated solids. This contributes to achieve one of China's stated sustainable development priorities "Actively adapt to climate change and strengthen resistance capacity to climate risks in agriculture, forestry, water resources and other key fields, as well as cities, coastal regions and ecologically vulnerable areas".

1.18 Additional Information Relevant to the Project

⁷ <https://sdgs.un.org/goals/goal12>

⁸ <https://sdgs.un.org/goals/goal13>

Leakage Management

Not applicable. Please refer to Section 5.5 for details.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

Not applicable.

2 SAFEGUARDS

2.1 No Net Harm

The Environment Impact Assessment (EIA) for the project was approved by Ecology and Environment Bureau of Hengyang City on 15-Jan-2021 (Heng Huan Lei Ping [2021] No.15). This EIA report has assessed every possible aspect of environmental impact of the project and recommended corresponding mitigation measures, where applicable. No net harm has been detected. The project would also contribute to the sustainable development of local community as described in Section 1.17 above.

To conclude, the project would not bring negative impacts to local environment and socio-economy. No net harm on local environment and social community was detected for the project.

2.2 Local Stakeholder Consultation

A local stakeholder consultation (LSC) meeting was held at the office building on the project site in Jun-2020. During the meeting, project information and corresponding environmental and social impacts of the project were communicated thoroughly to the local stakeholders. 30 copies of questionnaire were distributed to stakeholder representatives, taking account of different ages, genders and occupations as shown in Table 2-1 below.

Table 2-1 Structure of stakeholder survey for the project

Category		Quantity	Percentage
Amount of stakeholders surveyed	Male	20	67%
	Female	10	33%
Age	<25	2	7%

	25-55	21	70%
	>55	7	23%
Education	Junior high school or below	22	73%
	Senior high school	6	20%
	College or above	2	7%
Occupation	Worker	8	27%
	Peasant	10	33%
	Management personnel	3	10%
	Civil servant	3	10%
	Unspecified	6	20%

Table 2-2 below summarizes the questions and responses received. Most local stakeholder representatives believe that the project will improve local people's lives, promote local economic development, and have little impact on the environment. All stakeholder representatives are supportive to the project activities. Therefore, the implementation of the project is considered beneficial.

Table 2-2 Summary of stakeholders' comments on the project

No.	Questions	Attitude or Opinion	Amount	Percentage
1	Do you know about the project activity?	Very much	19	63%
		Heard of	8	27%
		Nothing	3	10%
2	Do you think the project will improve the natural environment?	Yes	27	90%
		No	0	0%
		Don't know	3	10%
3	Do you think the project will improve the local employment situation?	Yes	18	60%
		No	6	20%
		Don't know	6	20%
4	Do you think the project will improve the local social community?	Yes	21	70%
		No	3	10%
		Don't know	6	20%

5	What is the most probable environmental impact do you think the project will cause after implementation of the project?	None	29	97%
		Air pollution	0	0%
		Water pollution	0	0%
		Noise pollution	0	0%
		Harm to land	0	0%
		Don't know	1	3%
6	What is your attitude to the project activity?	Support	30	100%
		Against	0	0%
		Indifferent	0	0%

The ongoing communications with local stakeholders will be carried out at periodic intervals, where key implementation schedules or changes of the project are conveyed to local authorities, neighbourhood committees, and residents. Comments and suggestions from any of these entities are collected and considered by the project owner.

2.3 Environmental Impact

The EIA report of the project was approved by Ecology and Environment Bureau of Hengyang City on 15-Jan-2021 (Heng Huan Lei Ping [2021] No.15). The environmental impacts of the project operation are summarized as follows.

1) Wastewater

The production wastewater of this project includes wastewater from equipment and vehicle cleaning. This is collected into the pre-treatment tank for fertilizer production without being discharged.

2) Air pollution

● Odorous gas

The project builds a fully enclosed pre-treatment tank to temporarily store the wastes, and solid separation and composting take place in the solid fertilizer workshop equipped with ventilation facilities. Sewage collection and transportation systems take place in an enclosed environment. Greenspace with trees and shrubs are settled surrounding the production workshops. The collected odorous gas is purified by the biofilter tower and discharged through the 15m exhaust cylinder. These result in a minimal influence of odorous gas on the atmospheric environment.

● Dust

The project generates dust during the crushing, screening and drying of manure solids. Waste gas from crushing and screening is collected and treated by the bag dust collectors and

discharged into the air by a 15m exhaust cylinder. Waste gas from drying is collected and treated by a built-in spiral sinking bag dust collector and recycled in the solid fertilizer workshop.

3) Noise

The noise during the operation of the project mainly comes from transport vehicles. The transportation volume and time period are strictly regulated, no transport vehicle will enter the project site during nighttime (22:00 - 6:00).

4) Solid waste

The project does not generate solid wastes during the solid separation and composting procedures.

2.4 Public Comments

The project will be open for public comment on the VERRA website. The project shall be listed, and comments shall be incorporated later.

2.5 AFOLU-Specific Safeguards

Not applicable.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The project follows the VSC Standard version 4.5 and uses the following methodologies and tools.

- CDM Small-scale Methodology AMS-III.Y: "Methane avoidance through separation of solids from wastewater or manure treatment systems", version 04.0
<https://cdm.unfccc.int/methodologies/DB/IR1ULTHWQKPQ099203UJTLELME23L>
- CDM Small-scale Methodology AMS-III.F: "Avoidance of methane emissions through composting", version 12.0
<https://cdm.unfccc.int/methodologies/DB/NZ83KB7YHBIA7HL2U1PCNAOCHPUQYX>
- CDM Small-scale Methodology AMS-I.D: "Grid connected renewable electricity generation", version 18.0
<https://cdm.unfccc.int/methodologies/DB/W3TINZ7KKWCK7L8WTXFQ00FQ0H4SBK>
- CDM Methodological Tool 05: "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation", version 03.0

https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-05-v3.0.pdf/history_view

- CDM Methodological Tool 13: “Project and leakage emissions from composting”, version 02.0

https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-13-v2.pdf/history_view

- CDM Methodological Tool 21: “Demonstration of additionality of small-scale project activities”, version 13.1

https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-21-v13.1.pdf/history_view

3.2 Applicability of Methodology

The project meets the applicability criteria outlined in the methodology and associated tools in the following ways.

- CDM Small-scale Methodology AMS-III.Y: “Methane avoidance through separation of solids from wastewater or manure treatment systems”, version 04.0:

Para.	Criteria	Justification for the Project Activity
7	In case of animal manure management systems the following conditions apply: a) Animals shall be managed in confined conditions; b) If organic bedding material is used in the animal barns or added to the manure stream, separated solids shall not be reused in the barns; c) If the baseline manure slurry was treated in an anaerobic lagoon or another liquid treatment system, the outflow liquid from the lagoon was recycled as flush water or used to irrigate fields; however it was not discharged into river/lake/sea. In the latter case, i.e. effluent discharge into river/lake/sea, the system is considered as a wastewater treatment	Applicable. The project involves animal manure management systems with the following conditions applied: a) The project serves for local farms that have a high productivity, where animals are managed in confined conditions; b) The project utilizes the separated solids to produce organic fertilizers through composting and will not reuse separated solids in the barns; c) The baseline manure slurry was treated in anaerobic lagoons and the outflow liquid from the lagoon was used to irrigate fields;

	system and not a manure management system; d) A minimum interval of six months was observed between each removal of the solids accumulated in the lagoon.	d) A minimum interval of six months was observed between each removal of the solids accumulated in the lagoons.
8	In case of wastewater treatment systems the following conditions apply: a) The baseline treatment systems do not include a fine solids separation process (i.e. grading smaller than 10 mm aperture, primary settlers, mechanical separation, etc.);⁹ b) In case the baseline treatment system was an anaerobic lagoon or a liquid system, a minimum interval of 30 days was observed between each removal of the solids accumulated in the lagoon.	Irrelevant. There is no water treatment system involved in this project.
9	This methodology is not applicable when the project treats solids removed from an existing lagoon, or sludge originated from settlers or from any other biologically active treatment device of the baseline animal manure management/wastewater treatment system.	Irrelevant. The project does not treat solids removed from an existing lagoon, or sludge originated from settlers or from any other biologically active treatment device.

⁹ This is to exclude the situation where in the absence of the project activity, the COD content of the solids separated in baseline would not have decayed in the wastewater treatment system, but in a solid waste disposal site or in any other solid waste management system. However, it shall be distinguished from the situation where certain solid separation equipment are used in the baseline to serve as a necessary/standard step in the industrial and manufacturing process, and continue to be used during the project activity. For example, filter press is commonly used in alcohol industry to recover the fine solids with an economic value. In such a case, this provision will not apply, because only the effluent after passing the filter press will be considered as wastewater.

10	The separated solids shall be further treated and methane emissions resulting from further treatment, storage, use or disposal shall be considered. If the solids are combusted for thermal or heat generation, that component of the project activity can use a corresponding methodology under Type I. If the solids are mechanically/thermally treated to produce refuse-derived fuel (RDF) or stabilized biomass (SB) the relevant provisions in AMS-III.E shall be followed. If the solids are used as animal feeds (e.g. feed to cows, pigs), emissions from enteric fermentation and emissions from the manure may be neglected.	Applicable. The separated solids are further composted to produce organic solid fertilizers, and the resulting methane emissions are included in the project emissions.
11	The liquid fraction from the project solid separation system shall be treated either in the baseline treatment facility or in a treatment system with a lower methane conversion factor (MCF) than the baseline system.	Applicable. The liquid fraction from the project solid separation system is diverted to produce liquid fertilizers in a water management system that combines aerobic and low-leakage anaerobic treatments, which has a lower MCF than the baseline uncovered anaerobic lagoons.
12	This methodology applies to situations where the baseline treatment systems have been operational at least three years before the start date of the project activity. New facilities (Greenfield projects) and project activities involving a change of equipment resulting in an efficiency improvement or capacity addition of the wastewater or sludge treatment system compared to the designed capacity of the baseline treatment system are only eligible to apply this methodology if they comply with the requirements in the “General Guidelines to SSC CDM methodologies” concerning these topics. In addition, the	Applicable. The baseline situation where the solids separated being left in uncovered anaerobic lagoons was a long-term (at least 3 years) common practice to local farmers, until the presence of this project.

	requirements for demonstration of the remaining lifetime of the equipment replaced as described in the general guidelines shall be followed.	
13	In case flocculants is used in the project activities, project emissions and leakage from use of flocculants should be taken into account.	Irrelevant. The project does not use flocculants in the project activity.
14	Measures are limited to those that result in emission reductions of less than or equal to 60 Kt CO ₂ equivalent annually.	Applicable. The annual emission reductions of the project do not exceed 60 Kt CO ₂ .

- CDM Small-scale Methodology AMS-III.F: “Avoidance of methane emissions through composting”, version 12.0: The primary methodology AMS-III.Y directs the project to use this methodology to quantify project emissions from composting of solid separated.
- CDM Small-scale Methodology AMS-I.D: “Grid connected renewable electricity generation”, version 18.0: The primary methodology AMS-III.Y directs the project to use this methodology to quantify project emissions from electricity consumption of solid separation.
- CDM Methodological Tool 13: “Project and leakage emissions from composting”, version 02.0. The methodology AMS-III.F directs the project to use this tool to quantify project emissions from composting of solid separated.

Para.	Criteria	Justification for the Project Activity
3	The following sources of project emissions are accounted for in this tool: a) CH ₄ and N ₂ O emission from composting; b) CO ₂ emissions from consumption of fossil fuels and electricity associated with composting; and c) CH ₄ emissions from run-off wastewater associated with co-composting.	Applicable. a) CH ₄ and N ₂ O emissions from composting are taken into consideration; b) CO ₂ emissions from consumption of fossil fuels and electricity associated with composting are taken into consideration; c) CH ₄ emissions from run-off wastewater associated with co-composting do not exit.
4	The following source of leakage emissions is accounted for in this tool:	Irrelevant. The residual organic content of compost is neither disposed of in a landfill nor

	a) CH ₄ emissions from the anaerobic decay of the residual organic content of compost disposed of in a landfill or subjected to anaerobic storage.	subjected to anaerobic storage, thus the composting procedure does not generate CH ₄ emissions from the anaerobic decay.
5	Transport emissions are not accounted for in this tool because it is assumed that similar transportation activities would occur in the baseline.	Applicable. Transport emissions are not taken into consideration.

- CDM Methodological Tool 05: “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation”, version 03.0. The methodology AMS-III.F directs the project to use this tool to quantify project emissions from electricity consumption by composting of solid separated.

Para.	Criteria	Justification for the Project Activity
5	If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption: a) Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or, if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer. b) Scenario B: Electricity consumption from (an) off-grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or	Applicable. The project calculates emissions from electricity consumption. Scenario A is applicable as the project consumes electricity from the local power grid connected to the Central China Power Grid (CCPG).

	c) Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants operate at the site of the electricity consumer. The captive power plant(s) can provide electricity to the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity consumer can be provided with electricity from the captive power plant(s) and the grid	
6	This tool can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, only if one out of the following three project scenarios applies to the recipient of the electricity generated: a) Scenario I: Electricity is supplied to the grid; b) Scenario II: Electricity is supplied to consumers/electricity consuming facilities; or c) Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities	Irrelevant. The project does not involve electricity generation in the project scenario.
7	This tool is not applicable in cases where captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. The tool only accounts for CO₂ emissions.	Irrelevant. The project does not involve installation of captive renewable power generation technologies.

- CDM Methodological Tool 21: “Demonstration of additionality of small-scale project activities”, version 13.1:

Para.	Criteria	Justification for the Project Activity
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4	The use of the methodological tool “Demonstration of additionality of small-scale project activities” is not mandatory for project participants when proposing new methodologies. Project participants and coordinating/managing entities may propose alternative methods to demonstrate additionality for consideration by the Executive Board.	Irrelevant. The PP will not propose any new methodologies and will not propose any alternative methods to demonstrate additionality.
5	Project participants and coordinating/managing entities may also apply “TOOL19: Demonstration of additionality of microscale project activities” as applicable.	Irrelevant. The project is a small-scale project as the annual emission reductions of the project does not exceed 60 Kt CO ₂ threshold designated by CDM.

3.3 Project Boundary

As per para.20 of AMS-III.Y, the project boundary applicable to the proposed project activity is the physical, geographical site of:

- 1) Local swine farms and uncovered anaerobic lagoons where the animal waste would have been collected, stored and treated and the methane emission would have occurred in the absence of the proposed project activity;
- 2) The project site where the treatment of animal waste through solids separation takes place;
- 3) The project site where the composting of the separated solids takes place;
- 4) The itineraries between where the transportation of separated solids occur.

Source	Gas	Included?	Justification/Explanation
Baseline	CO ₂	No	Excluded for simplification. This emission source is assumed to be very small
	CH ₄	Yes	The major emission source
	N ₂ O	No	Excluded for simplification. This emission source is assumed to be very small
	Other	No	Excluded for simplification. This emission source is assumed to be very small
Project	CO ₂	Yes	Involves emission from composting, which may be an important emission source

Source		Gas	Included?	Justification/Explanation
	destruction or disposal of solids separated	CH ₄	Yes	Involves emission from composting, which may be an important emission source
		N ₂ O	Yes	Involves emission from composting, which may be an important emission source
		Other	No	Excluded for simplification. This emission source is assumed to be very small
	Emissions from electricity and fossil fuel use by the solid separation technology	CO ₂	Yes	May be an important emission source
		CH ₄	No	Excluded for simplification. This emission source is assumed to be very small
		N ₂ O	No	Excluded for simplification. This emission source is assumed to be very small
		Other	No	Excluded for simplification. This emission source is assumed to be very small
	Emissions from the combustion of flocculants used in the pre-separation phase	CO ₂	No	Does not exist
		CH ₄	No	Does not exist
		N ₂ O	No	Does not exist
		Other	No	Does not exist
	Incremental CO ₂ emissions due to increased transportation	CO ₂	No	Does not exist
		CH ₄	No	Does not exist
		N ₂ O	No	Does not exist
		Other	No	Does not exist

Figure 3-1 below further depicts the project boundary and activities.

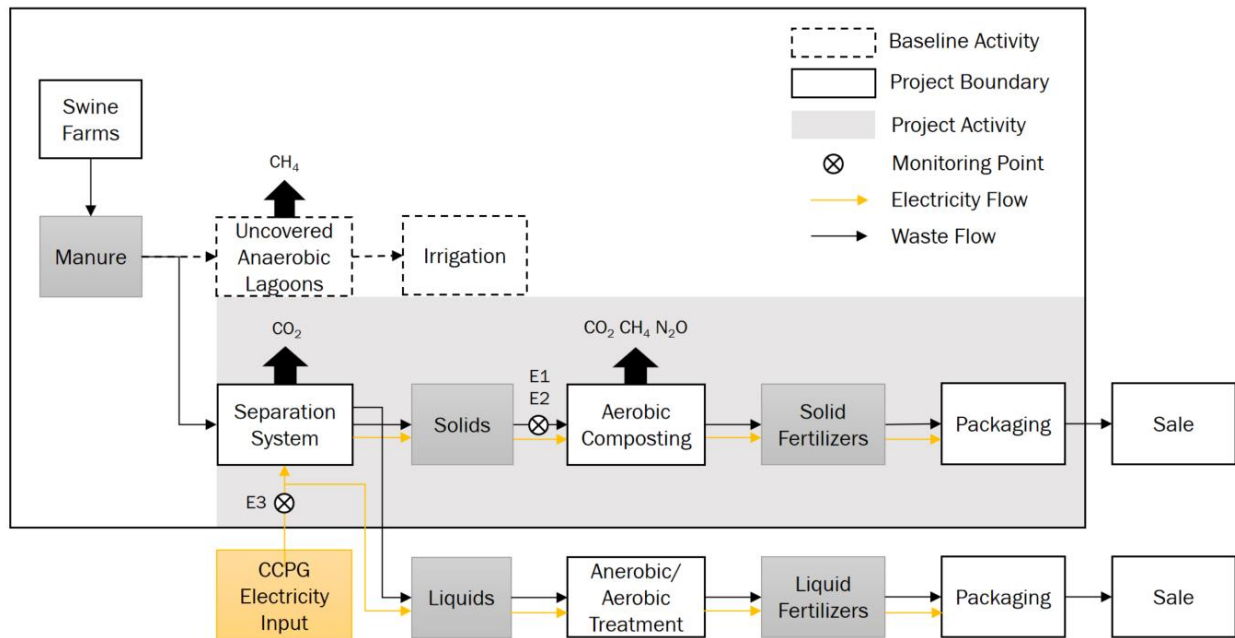


Figure 3-1 Project Boundary

3.4 Baseline Scenario

In the baseline scenario of this project, manure from swine farms was left to decay in uncovered lagoons for uncontrolled anaerobic decay and was used seasonally for irrigation. The lagoons were barely fully drained and cleaned, and the methane produced was not recovered and was emitted directly to the atmosphere.

3.5 Additionality

The project activity, which introduces an animal manure composting system associated with solid separation techniques to avoid methane emissions through the separation of manure solids, is not mandated by any of the identified regulatory frameworks listed in Section 1.14.

As per para.17 of AMS-III.Y, the PP shall take into account the general guidance to the small-scale (SSC) CDM methodologies and the Tool for demonstration of additionality of SSC project activities. The project therefore applies Tool 21 to demonstrate additionality due to its size, which does not exceed the small-scale CDM threshold of 60 Kt CO_2 .

Tool 21 requires the PP to demonstrate at least one obstacle included in the tool to show that the project activity would not have occurred anyway. The investment barrier was identified as the most relevant barrier.

The PP conducted a benchmark analysis based on the internal rate of return (IRR) of project cash flows to assess the financial position of the proposed project activities in the absence of the

revenue from carbon credit. The revenue source of the project is the production and sales of organic fertilizers, which is classified into the fertilizer industry according to the "Industrial Classification for National Economic Activities" (GB/T 4754-2017)¹⁰. Detailed procedures of the benchmark analysis are demonstrated below.

1) IRR calculation

The calculation results of the project's IRR is 5.70%.

2) Comparing project's IRR with post-tax equity IRR benchmark for the fertilizer industry

According to the "Table of the Financial Benchmark Rate of Return for Project Constructions" issued by the National Development and Reform Commission of China, the post-tax equity IRR benchmark for the fertilizer industry is 10%¹¹. The project's IRR of 5.70% is much lower than the post-tax equity IRR, suggesting that the proposed project is financially unacceptable due to such a low profitability and thus investment barrier occurs.

3) Sensitivity analysis

The sensitivity analysis is conducted to confirm whether, under reasonable deviations in the four financial indicators stated below, the project's IRR would not exceed the post-tax equity IRR without receiving revenue from carbon credit, and thus would still remain financially unattractive. The four financial indicators that significantly affect the project's IRR are:

- i. Static investment
- ii. Annual organic fertilizer production
- iii. Annual O&M cost
- iv. Price of organic fertilizer

The impacts of the above indicators on the project's IRR were analyzed. The results of this sensitivity analysis are shown in Table 3-2.

Table 3-2 Sensitivity analysis of the project's IRR to different financial indicators

Hypothetical change	-10%	-5%	0	+5%	+10%
Static investment	7.39%	6.51%	5.70%	4.96%	4.28%
Annual organic fertilizer production	1.08%	3.45%	5.70%	7.87%	9.96%
Annual O&M cost	8.77%	7.25%	5.70%	4.11%	2.46%

¹⁰ <https://max.book118.com/html/2021/1017/8140033046004021.shtm>

¹¹ https://www.ndrc.gov.cn/fggz/gdzctz/tzfg/201907/t20190729_1197578.html?code=&state=123

Price of organic fertilizer	1.08%	3.45%	5.70%	7.87%	9.96%
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4) Conclusion

As evidenced by the financial statistics presented herein, the project faces an investment barrier and is therefore considered additional.

3.6 Methodology Deviations

Not applicable.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

In the case of manure stream where the barns do not use any organic bedding materials, baseline emissions are calculated based on the total mass of the volatile solids separated as per para.23 of AMS-III.Y:

$$BE_y = (B_{o,w,y} \times M_{ss,y} \times VS_{ss,y} \times UF_b \times GWP_{CH_4} \times D_{CH_4}/1000) \times \sum_i (MS_{BL,i} \times MCF_{b,i}) \quad \text{Equation (1)}$$

Where:

BE_y	=	Baseline emissions in year y (tCO ₂ e)
$B_{o,w,y}$	=	Weighted methane-producing potential of the volatile solids separated by the project in year y (m ³ CH ₄ per kg of VS). The project only involves manure from swine.
$M_{ss,y}$	=	Mass (dry matter basis) of total separated solids in year y (kg)
$VS_{ss,y}$	=	Volatile solids content of the separated solids in year y on a dry matter basis (kg/kg)
UF_b	=	Model correction factor to account for model uncertainties (0.94) ¹²

¹² Reference: FCCC/SBSTA/2003/10/Add.2, page 25

GWP_{CH_4}	=	Global Warming Potential of methane applicable to the crediting period (t CO ₂ e/tCH ₄)
D_{CH_4}	=	Conversion factor of m ³ CH ₄ to kilograms (0.67 kg per m ³ at 20°C and 1 atm pressure)
i	=	Index for baseline anaerobic manure management system
$MS_{BL,i}$	=	Fraction of manure handled in the baseline anaerobic manure management system i (fraction, mass basis), based on historical information from previous year(s) before project. If historical information is not available, the capacity (t/day) of each baseline management system shall be verified before the project start. During the project, it would be assumed that the management systems with lower MCFs would be used up to their full capacity, and only thereafter the systems with larger MCF would be used. The value of $MS_{BL,i}$ is “1.0” if only one type of treatment system was used in the baseline
$MCF_{b,i}$	=	Methane conversion factor for the baseline anaerobic manure management system i as per IPCC Guidelines for National Greenhouse Gas Inventories

4.2 Project Emissions

Project emissions are quantified according to AMS-III.Y and AMS-III.F.

According to para.29 of AMS-III.Y, project activity emissions consist of:

- 1) Any methane emissions from storage, use, disposal or destruction of solids separated ($PE_{y,ss}$);
- 2) Emissions from electricity and fossil fuel use by the solid separation technology as per the methods of AMS-I.D ($PE_{y,power}$);
- 3) Emissions from the combustion of flocculants used in the pre-separation phase ($PE_{y,floc,combustion}$);
- 4) Incremental CO₂ emissions due to increased transportation ($PE_{y,transp}$):
 - i. Transportation of solids to sites where it will be treated further or gainfully used (within the project boundary);
 - ii. Transportation of solids from and to treatment facilities to storage sites (within the project boundary);

iii. Transportation of solids to disposal site.

$$PE_y = PE_{y,ss} + PE_{y,power} + PE_{y,floc,combustion} + PE_{y,transp} \quad \text{Equation (2)}$$

Where:

$PE_{y,ss}$ = Project emissions from storage, use, destruction or disposal of solids separated in year y (tCO₂e).

$PE_{y,power}$ = Project emissions from energy use for operating the separation device in year y (tCO₂e), calculated as per AMS-I.D methods

$PE_{y,floc,combustion}$ = Project emissions from combustion of flocculants in year y (tCO₂e)

$PE_{y,transp}$ = Project emissions for incremental transportation of solids in the project scenario, beyond the emissions for transportation of solids in the baseline scenario (tCO₂e)

4.2.1 Project emissions from storage, use, destruction or disposal of solids separated ($PE_{y,ss}$)

$PE_{y,ss}$ is the project emissions from storage, use, destruction or disposal of solids separated in year y (tCO₂e). The project does not store the separated solids on-site and no solids are disposed either. The collected swine manure is all put into fertilizer production through composting right after being separated. Therefore, this section only takes into account the project emissions from the use of solids separated ($PE_{s,composting,y}$), which is composting, and so:

$$PE_{y,ss} = PE_{s,composting,y} \quad \text{Equation (3)}$$

According to para.34 of AMS-III.Y, in case the solids separated from manure or from wastewater are composted, the methane emissions from composting of solids separated by the project activity ($PE_{s,composting,y}$) is determined as per the relevant procedure in AMS-III.F. According to para.26 of AMS-III.F, $PE_{s,composting,y}$ is determined as per Section 6.1 of Tool 13:

$$PE_{s,composting,y} = PE_{EC,y} + PE_{FC,y} + PE_{CH_4,y} + PE_{N_2O,y} + PE_{RO,y} \quad \text{Equation (4)}$$

Where:

$PE_{s,composting,y}$ = Project emissions in the year y (tCO₂e)

$PE_{EC,y}$ = Project emissions from electricity consumption associated with composting in year y (tCO₂/yr)

$PE_{FC,y}$ = Project emissions from fossil fuel consumption associated with composting in year y (tCO₂/yr)

$PE_{CH_4,y}$	=	Project emissions of methane from the composting process in year y (tCO ₂ e/yr)
$PE_{N_2O,y}$	=	Project emissions of nitrous oxide from the composting process in year y (tCO ₂ e/yr)
$PE_{RO,y}$	=	Project emissions of methane from run-off wastewater associated with co-composting in year y (tCO ₂ e/yr)

1) Determination of the quantity of waste composted (Q_y)

Q_y is the quantity of waste composted in year y (t/yr), which is a parameter required in the determination of emissions associated with each source of project emissions. It is determined by monitoring the weight of separated solids delivered to the composting installation using an on-site weighing device as per Option 1, Section 6.1.1.2 of Tool 13.

2) Determination of composting emissions from electricity consumption ($PE_{EC,y}$)

$PE_{EC,y}$ is the project emissions from electricity consumption associated with composting in year y (tCO₂/yr). For the project, the composting activity involves electricity consumption from electricity-driven devices for production use. As part of the project activity, composting takes place at the project site, where the electricity consumed by composting, solid separation and all related facilities within the project boundaries are measured as a whole. To avoid underestimating the project emissions, the electricity consumption of the composting activity is then determined based on the total electricity consumption within the project boundary. This is intended to calculate the quantity of electricity consumption in a more accurate way by relying on continuously measures rather than non-monitored parameters. Therefore, as per para. 17 of Tool 13, $PE_{EC,y}$ is calculated using the latest approved version of Tool 05. As per Section 6.2.1 of Tool 05, composting emissions from consumption of electricity is calculated as follows:

$$PE_{EC,y} = \sum_j EC_{PJ,compost,y} \times EF_{EF,compost,y} \times (1 + TDL_{compost,y}) \quad \text{Equation (5)}$$

Where:

$PE_{EC,y}$	=	Composting emissions from electricity consumption in year y (tCO ₂ /yr)
$EC_{PJ,compost,y}$	=	Quantity of electricity consumed by composting in year y (MWh/yr)
$EF_{EF,compost,y}$	=	Emission factor for electricity generation for composting in year y (tCO ₂ /MWh). Default value of 1.3 tCO ₂ /MWh suggested by Tool 05 shall apply
$TDL_{compost,y}$	=	Average technical transmission and distribution losses for providing electricity to composting in year y. Default value of 20% suggested by Tool 05 shall apply

As stated above, to be conservative, the electricity consumption within the project boundary ($EC_{PJ,project,y}$) is used to determine the electricity consumption emissions caused by composting ($EC_{PJ,compost,y}$), and so:

$$EC_{PJ,compost,y} = EC_{PJ,project,y} \quad \text{Equation (6)}$$

3) Determination of composting emissions from fossil fuel consumption ($PE_{FC,y}$)

$PE_{FC,y}$ is the project emissions from fossil fuel consumption associated with composting in year y (tCO_2/yr). For the project, the composting activity involves fossil fuel consumption from fuel vehicles for transportation and production use. As per Option 2 suggested by Section 6.1.3 of Tool 13, it is calculated as follows:

$$PE_{FC,y} = Q_y \times EF_{FC,default} \quad \text{Equation (7)}$$

Where:

$PE_{FC,y}$ = Project emissions from fossil fuel consumption associated with composting in year y (tCO_2/yr)

$EF_{FC,default}$ = Default emission factor for fossil fuels consumed by the composting activity per tonne of waste (tCO_2/t)

4) Determination of composting emissions of methane ($PE_{CH_4,y}$)

$PE_{CH_4,y}$ is the project emissions of methane from the composting process in year y (tCO_2e/yr). As per Section 6.1.4 of Tool 13, it is calculated as follows:

$$PE_{CH_4,y} = Q_y \times EF_{CH_4,y} \times GWP_{CH_4} \quad \text{Equation (8)}$$

Where:

$PE_{CH_4,y}$ = Project emissions of methane from the composting process in year y (tCO_2e/yr)

$EF_{CH_4,y}$ = Emission factor of methane per tonne of waste composted valid for year y (tCH_4/t).

GWP_{CH_4} = Global Warming Potential of CH_4 (tCO_2e/tCH_4)

5) Determination of composting emissions of nitrous oxide ($PE_{N_2O,y}$)

$PE_{N_2O,y}$ is the project emissions of nitrous oxide from the composting process in year y (tCO_2e/yr). As per Section 6.1.5 of Tool 13, it is calculated as follows:

$$PE_{N_2O,y} = Q_y \times EF_{N_2O,y} \times GWP_{N_2O} \quad \text{Equation (9)}$$

Where:

$PE_{N2O,y}$ = Project emissions of nitrous oxide from composting in year y (tCO_{2e}/yr)

$EF_{N2O,y}$ = Emission factor of nitrous oxide per tonne of waste composted valid for year y (tN₂O/t). This is determined following Option 2: Procedure using default values as per Section 6.1.5.2 of Tool 13, where $EF_{N2O,y} = EF_{N2O,default}$

GWP_{N2O} = Global Warming Potential of N₂O (tCO_{2e}/tN₂O)

6) Determination of composting emissions from run-off wastewater ($PE_{RO,y}$)

$PE_{RO,y}$ is the project emissions of methane from run-off wastewater associated with co-composting in year y (tCO_{2e}/yr). The project does not involve co-composting activities, thus $PE_{RO,y}$ is zero.

4.2.2 Project emissions from energy use for operating separation devices ($PE_{y,power}$)

$PE_{y,power}$ is the project emissions from energy use for operating the separation device in year y (tCO_{2e}), and is calculated as follows:

$$PE_{y,power} = PE_{y,electricity} + PE_{y,fuel} \quad \text{Equation (10)}$$

According to para.35 of AMS-III.Y, project activity emissions from electricity consumption are determined as per the procedures described in AMS-I.D. The energy consumption of all equipment/devices installed by the project activity, inter alia all equipment to separate solids (including energy used for spray drying and evaporation) shall be included. AMS-I.D calculates CO₂ emissions as follows:

$$PE_{y,electricity} = EG_{PJ,y} \times EF_{grid,y} \quad \text{Equation (11)}$$

For the project, the entire solid separation system is electricity-driven, and the electricity consumed is measured together with other facilities within the project boundary. Since Section 4.2.1 has already calculated the emission from total electricity consumption within the project boundary using Equation (5) to avoid underestimating composting emissions, $PE_{y,electricity}$ here is therefore considered to be zero to avoid double counting.

This does not underestimate the total electricity consumption within the project boundary, as Equation (5) further considers the average technical transmission and distribution losses for providing electricity to the project (TDL), leading to a more conservative result of project emissions than that of Equation (11).

It is also worth noting that the electricity demand for the project is all supplied by the local power grid that is connected to the Central China Power Grid (CCPG), where the historical emission factors for the recent years are consistently lower than 1.0 tCO₂/MWh. As the default emission factor of 1.3 tCO₂/MWh suggested by Tool 05 is used to calculate $PE_{EC,y}$ using Equation (5), it in fact assumes a more conservative estimation on project emissions.

There is no additional fossil fuel combustion, thus $PE_{y,fuel}$ is zero.

4.2.3 Project emissions from combustion of flocculants ($PE_{y,floc,combustion}$)

$PE_{y,floc,combustion}$ is the project emissions from combustion of flocculants in year y (tCO_2e). According to para.36 of AMS-III.Y, in case of combustion of separated solids, the CO_2 emissions from the combustion of the non-biomass (i.e. fossil) carbon content of the flocculants shall be considered. In such a case, a default value of $0.2 tCO_2e$ per tonne of flocculants may be used.

The project does not use flocculants during solid separations, thus $PE_{y,floc,combustion}$ is zero.

4.2.4 Project emissions from incremental transportation of separated solids ($PE_{y,trans}$)

$PE_{y,trans}$ is the project emissions for incremental transportation of solids in the project scenario, beyond the emissions for transportation of solids in the baseline scenario (tCO_2e). According to para.37 of AMS-III.Y, the incremental transportation of separated solids for storage, gainful use, destruction and/or land application is calculated as follows:

$$PE_{y,transp} = Q_{y,transp} / CT_y \times DT_y \times EF_{CO_2} \quad \text{Equation (12)}$$

Where:

$Q_{y,transp}$	=	Quantity of solids transported in year y (t)
CT_y	=	Average truck capacity for transportation (t/truck)
DT_y	=	Average incremental distance for transportation of separated solids (km/truck)
EF_{CO_2}	=	CO_2 emission factor for the fuel used for transportation (tCO_2/km , IPCC default values or local values)

For the project, all separated solids are composted on site prior to any secondary use, without storage, destruction or land application. The solids from the separation system immediately enter the composting system, and the transportation distance between these two systems is minimal and can be accomplished by human forces alone. There is no additional transportation, thus $PE_{y,transp}$ is zero.

4.3 Leakage

The project does not have any leakage emissions to be addressed, due to the following reasons:

- 1) Solid separation technologies of the project are brand new at the time of commissioning and are not transferred from another activity;

- 2) The project does not use flocculants during the project activity. However, if flocculants is used and if the flocculants includes ingredients that were manufactured and not a waste product, then leakage emissions must be addressed for that portion of the flocculants. According to para.39 of AMS-III.Y, a default emission factor of 7.9 tCO₂e/t may be applied based on the amount of the flocculants that is manufactured (i.e. excluding the portions that obtained from waste sources).

4.4 Net GHG Emission Reductions and Removals

Emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y - LE_y \quad \text{Equation (13)}$$

Where:

ER_y = Emission reductions during the year y (tCO₂)

BE_y = Baseline emissions during the year y (tCO₂)

PE_y = Project emissions during the year y (tCO₂)

LE_y = Leakage during the year y (tCO₂)

Based on the calculation methods stated above, the ex-ante GHG emission reductions is calculated as follows:

$$BE_y = (B_{o,w,y} \times M_{ss,y} \times VS_{ss,y} \times UF_b \times GWP_{CH_4} \times D_{CH_4}/1000) \times \sum_i (MS_{Bl,i} \times MCF_{b,i})$$

Parameter	Ex-ante Value	Unit	Sources
$B_{o,w,y}$	0.45	m ³ CH ₄ /kg-VS	Default, IPCC 2019 refinement, Table 10.16A, Swine, Other Regions, High productivity systems
$M_{ss,y}$	21,000,000	kg	Feasibility Study Report
$VS_{ss,y}$	0.55	kg/kg	Feasibility Study Report
UF_b	0.94	-	Default, AMS-III.Y
GWP_{CH_4}	28.0	tCO ₂ /tCH ₄	Default, IPCC AR5
D_{CH_4}	0.67	kg/m ³	Default, AMS-III.Y
$MS_{Bl,i}$	100%	-	Default, AMS-III.Y

MCF _{b,i}	73%	-	Default, IPCC 2019 refinement, Table 10.17, Uncovered anaerobic lagoon, Warm Temperate Moist
BE _y	66,907	tCO ₂ e	Calculated

$$PE_y = PE_{y,ss} + PE_{y,power} + PE_{y,floc,combustion} + PE_{y,transp}$$

Parameter	Ex-ante Value	Unit	Sources
PE _{y,ss}	6,966	tCO ₂ e	Calculated
PE _{y,power}	-	tCO ₂ e	-
PE _{y,floc, combustion}	-	tCO ₂ e	-
PE _{y,transp}	-	tCO ₂ e	-
PE _y	6,966	tCO ₂ e	Calculated

$$PE_{s,composting,y} = PE_{EC,y} + PE_{FC,y} + PE_{CH_4,y} + PE_{N_2O,y} + PE_{R_0,y}$$

Parameter	Ex-ante Value	Unit	Sources
PE _{EC,y}	156.00	tCO ₂ e	Calculated
PE _{FC,y}	1,086.75	tCO ₂ e	Calculated
PE _{CH₄,y}	2,940.00	tCO ₂ e	Calculated
PE _{N₂O,y}	2,782.50	tCO ₂ e	Calculated
PE _{R₀,y}	-	tCO ₂ e	Irrelevant
PE _{s,composting,y}	6,966	tCO ₂ e	Calculated

$$PE_{EC,y} = \sum_j EC_{PJ,compost,y} \times EF_{EF,compost,y} \times (1 + TDL_{compost,y})$$

Parameter	Ex-ante Value	Unit	Sources
EC _{PJ,j,y}	100	MWh/y	Feasibility Study Report
EF _{EF,j,y}	1.3	tCO ₂ e/MWh	Default, Tool 5
TDL _{j,y}	20%	-	Default, Tool 5
PE _{EC,y}	156.00	tCO ₂ e/yr	Calculated

$$PE_{FC,y} = Q_y \times EF_{FC,default}$$

Parameter	Ex-ante Value	Unit	Sources
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$EF_{CH_4, default}$	0.0207	tN ₂ O/t	Default, Tool 13
Q_y	52,500	t/y	Feasibility Study Report
$PE_{CH_4, y}$	1,086.75	tCO ₂ e/yr	Calculated

$$PE_{CH_4, y} = Q_y \times EF_{CH_4, y} \times GWP_{CH_4}$$

Parameter	Ex-ante Value	Unit	Sources
$EF_{CH_4, default}$	0.002	tN ₂ O/t	Default, Tool 13
GWP_{CH_4}	28	tCO ₂ /tCH ₄	Default, IPCC AR5
Q_y	52,500	t/y	Feasibility Study Report
$PE_{CH_4, y}$	2,940.00	tCO ₂ e/yr	Calculated

$$PE_{N_2O, y} = Q_y \times EF_{N_2O, y} \times GWP_{N_2O}$$

Parameter	Ex-ante Value	Unit	Sources
$EF_{N_2O, default}$	0.0002	tCH ₄ /t	Default, Tool 13
GWP_{N_2O}	265	tCO ₂ /tN ₂ O	Default, IPCC AR5
Q_y	52,500	t/y	Feasibility Study Report
$PE_{N_2O, y}$	2,782.50	tCO ₂ e/yr	Calculated

The resulting ex-ante estimate of emission reductions is shown in the table below. Since the crediting period begins and ends part-way through 2022 and 2029, these years' total reductions are adjusted (10 months of credit in 2022, 3 months of credit in 2029).

$$ER_y = BE_y - PE_y - LE_y$$

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
28-Mar-2022 to 31-Dec-2022	51,143	5,325	-	45,818
01-Jan-2023 to 31-Dec-2023	66,907	6,966	-	59,941
01-Jan-2024 to 31-Dec-2024	66,907	6,966	-	59,941
01-Jan-2025 to 31-Dec-2025	66,907	6,966	-	59,941
01-Jan-2026 to 31-Dec-2026	66,907	6,966	-	59,941

01-Jan-2027 to 31-Dec-2027	66,907	6,966	-	59,941
01-Jan-2028 to 31-Dec-2028	67,091	6,966	-	59,941
01-Jan-2029 to 27-Mar-2029	15,764	1,641	-	14,123
Total	468,349	48,762	-	419,587

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	$B_{o,w,y}$
Data unit	m ³ CH ₄ per kg of VS
Description	Weighted methane-producing potential of the volatile solids separated by the project in year y
Source of data	2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value applied	0.45 for swine in high productivity systems
Justification of choice of data or description of measurement methods and procedures applied	The project only collects manure from swine. No country or regional specific value is available. Default value of Other Regions from Tables 10.16A of 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories is applied As the project collects manure majorly from local swine farms that aim at fulfilling national and export markets, the values of high productivity systems are applied for conservativeness
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	UF_b
Data unit	-
Description	Model correction factor to account for model uncertainties
Source of data	AMS-III.Y. (Reference: FCCC/SBSTA/2003/10/Add.2, page 25)
Value applied	0.94

Justification of choice of data or description of measurement methods and procedures applied	Default value
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	GWP _{CH4}
Data unit	tCO ₂ e/tCH ₄
Description	Global Warming Potential of methane applicable to the crediting period
Source of data	IPCC Fifth Assessment Report
Value applied	28
Justification of choice of data or description of measurement methods and procedures applied	Default value
Purpose of Data	Calculation of baseline and project emissions
Comments	Shall be updated according to any future revision to VCS standard by VERRA

Data / Parameter	GWP _{N2O}
Data unit	tCO ₂ e/tN ₂ O
Description	Global Warming Potential of nitrous oxide applicable to the crediting period
Source of data	IPCC Fifth Assessment Report
Value applied	265
Justification of choice of data or description of measurement methods and procedures applied	Default value
Purpose of Data	Calculation of project emissions

Comments	Shall be updated according to any future revision to VCS standard by VERRA
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Data / Parameter	D_{CH_4}
Data unit	kg per m ³ at 20°C and 1 atm pressure
Description	Conversion factor of m ³ CH ₄ to kilograms
Source of data	AMS-III.Y
Value applied	0.67
Justification of choice of data or description of measurement methods and procedures applied	Default value
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$MS_{Bl,i}$
Data unit	-
Description	Fraction of manure handled in the baseline anaerobic manure management system i (fraction, mass basis)
Source of data	AMS-III.Y
Value applied	1.0
Justification of choice of data or description of measurement methods and procedures applied	Values should be based on historical information from previous year(s) before project. If historical information is not available, the capacity (t/day) of each baseline management system shall be verified before the project start During the project, it would be assumed that the management systems with lower MCFs would be used up to their full capacity, and only thereafter the systems with larger MCF would be used. The value of $MS_{Bl,i}$ is “1.0” if only one type of treatment system was used in the baseline For the project, there was only one type of treatment system existed prior to the project implementation, where animal manure was left to decay in uncovered anaerobic lagoon at the local swine farms. Therefore, $MS_{Bl,i}$ is 1.0

Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$MCF_{b,i}$
Data unit	-
Description	Methane conversion factor for the baseline anaerobic manure management system
Source of data	2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value applied	73%
Justification of choice of data or description of measurement methods and procedures applied	No country or regional specific value is available. Default value from Table 10.17 of Chapter 10, Volume 4, 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories is applied The project falls within warm, temperate, moist climate zone based off Figure 10A.1 of the same chapter. There was only one type of treatment system existed prior to the project implementation, where animal manure was left to decay in uncovered anaerobic lagoons. According to Table 10.17, the corresponding annual MCF of uncovered anaerobic lagoons in this climate zone is 73%
Purpose of Data	Calculation of baseline emissions
Comments	Shall be updated according to any future revision to VCS standard by VERRA

Data / Parameter	$EF_{EF,compost,y}$
Data unit	tCO ₂ /MWh
Description	Emission factor for electricity generation for composting in year y
Source of data	Tool 05
Value applied	1.3
Justification of choice of data or description of	As per Tool 05, the determination of the emission factors for electricity generation ($EF_{EF,j/k/l}$) in the project scenario depends on which scenario (A, B or C), as described in Section 2.2, paragraph 5 of Tool 05 that applies to the source of electricity

measurement methods and procedures applied	consumption that would be displaced in the baseline by electricity generated in the project. This project is considered as Scenario A as it consumes electricity from the power grid, and Tool 05 suggests two options of calculation: 1) Option A1: Calculate the combined margin emission factor of the applicable electricity system, using the procedures in the latest approved version of the “Tool to calculate the emission factor for an electricity system” ($EF_{EF,j/k,l} = EF_{grid,CM,y}$); or 2) Option A2: Use the following conservative default values: a) A value of 1.3 tCO₂/MWh if: i. Scenario A applies only to project and/or leakage electricity consumption sources but not to baseline electricity consumption sources; or ii. Scenario A applies to: both baseline and project (and/or leakage) electricity consumption sources; and the electricity consumption of the project and leakage sources is greater than the electricity consumption of the baseline sources; b) A value of 0.4 tCO₂/MWh for electricity grids where hydro power plants constitute less than 50% of total grid generation in average of the five most recent years, or based on long-term averages for hydroelectricity production, and a value of 0.25 t CO₂/MWh for other electricity grids. These values can be used if: i. Scenario A applies only to baseline electricity consumption sources but not to project or leakage electricity consumption sources; or Scenario A applies to: both baseline and project (and/or leakage) electricity consumption sources; and the electricity consumption of the baseline sources is greater than the electricity consumption of the project and leakage sources. The project applies Option A2 to determine $EF_{EF,comp,y}$ using the conservative default values. As the project scenario applies only to project electricity consumption sources but not to baseline electricity consumption sources, $EF_{EF,comp,y} = 1.3 \text{ tCO}_2/\text{MWh}$.
Purpose of Data	Calculation of project emissions
Comments	-
Data / Parameter	TDL _{compost,y}
Data unit	-

Description	Average technical transmission and distribution losses for providing electricity to composting in year y
Source of data	Tool 05
Value applied	20%
Justification of choice of data or description of measurement methods and procedures applied	As per Tool 05, in case of other scenarios (scenario A and scenario C, cases C.I and C.III), choose one of the following options: 1) Use annual average value based on the most recent data available within the host country; 2) Use as default values of 20% for: a) project or leakage electricity consumption sources; b) baseline electricity consumption sources if the electricity consumption by all project and leakage electricity consumption sources to which scenario A or scenario C (cases C.I or C.III) applies is larger than the electricity consumption of all baseline electricity consumption sources to which scenario A or scenario C (cases C.I or C.III) applies; 3) Use as default values of 3% for: a) baseline electricity consumption sources; b) project and leakage electricity consumption sources if the electricity consumption by all project and leakage electricity consumption sources to which scenario A or scenario C (cases C.I or C.III) applies is smaller than the electricity consumption of all baseline electricity consumption sources to which scenario A or scenario C (cases C.I or C.III) applies No country or regional specific value is available. As the project scenario applies only to project electricity consumption sources, the default value of 20% is applied
Purpose of Data	Calculation of project emissions
Comments	-

Data / Parameter	$EF_{FC,default}$
Data unit	tCO ₂ /t
Description	Default emission factor for fossil fuel consumed by the composting activity per tonne of waste composted (wet basis)

Source of data	Tool 13. The emission factor was selected based on a review of fossil fuel consumption per tonne of waste composted in relevant validation reports of CDM projects using a conservative default emission factor for diesel (from the 2006 IPCC Guidelines)
Value applied	0.0207
Justification of choice of data or description of measurement methods and procedures applied	Default value
Purpose of Data	Calculation of project emissions
Comments	-

Data / Parameter	$EF_{CH_4,y}$
Data unit	tCH ₄ /t
Description	Emission factor of methane per tonne of waste composted valid for year y
Source of data	Tool 13. The emission factor was selected based on studying published results of emission measurements from composting facilities, literature reviews on the subject and published emission factors. Data from recent, high quality sources was analyzed and a value conservatively selected from the higher end of the range in results
Value applied	0.002
Justification of choice of data or description of measurement methods and procedures applied	As per Tool 13, there are two options which the PP may choose for determining $EF_{CH_4,y}$: - Option 1: Procedure using monitored data $EF_{CH_4,y}$ is determined based on measurements of the methane emissions during a composting cycle - Option 2: Procedure using default values A default value is used: $EF_{CH_4,y} = EF_{CH_4,default}$, which is the default emission factor of methane per tonne of waste composted (wet basis), as per “Data and parameters not monitored” section of Tool 13 The project chooses Option 2 for determining $EF_{CH_4,y}$
Purpose of Data	Calculation of project emissions

Comments	-
Data / Parameter	$EF_{N_2O,y}$
Data unit	tN ₂ O/t
Description	Emission factor of nitrous oxide per tonne of waste composted valid for year
Source of data	As per “Data and parameters not monitored” section of Tool 13, the emission factor was selected based on studying published results of emission measurements from composting facilities, literature reviews on the subject and published emission factors. Data from recent, high quality sources was analyzed and a value conservatively selected from the higher end of the range in results
Value applied	0.0002
Justification of choice of data or description of measurement methods and procedures applied	As per Tool 13, there are two options which the PP may choose for determining $EF_{N_2O,y}$: 3) Option 1: Procedure using monitored data $EF_{N_2O,y}$ is determined based on measurements of the methane emissions during a composting cycle 4) Option 2: Procedure using default values A default value is used: $EF_{N_2O,y} = EF_{N_2O,default}$, which is the default emission factor of nitrous oxide per tonne of waste composted (wet basis), as per “Data and parameters not monitored” section of Tool 13 This project chooses Option 2 for determining $EF_{N_2O,y}$
Purpose of Data	Calculation of project emissions
Comments	-

5.2 Data and Parameters Monitored

Data / Parameter	$M_{ss,y}$
Data unit	kg
Description	Mass (dry matter basis) of total separated solids in year y
Source of data	Ex-ante values could be retrieved from the Feasibility Study Report Ex-post values are from the measuring records

Description of measurement methods and procedures to be applied	The mass of total separated solids on the dry matter basis is determined based on the wet weight and the dry matter content of the separated solids The wet weight of all outputs from separation system are measured by direct weighing, and its moisture content is measured through representative sampling. The moisture content is further used to calculate the dry matter content of the separated solids. The staff record the wet weight and the moisture content to the measuring records
Frequency of monitoring/recording	Continuously
Value applied	Ex-ante: 21,000,000 (Wet weight:52,500 t; Moisture content: 60%)
Monitoring equipment	Weighing device, moisture meters
QA/QC procedures to be applied	Monitoring equipment installed on-site should be calibrated by an accredited organization according to national standards
Purpose of data	Calculation of baseline emissions
Calculation method	For the project, composting is the only way of using the separated solids and the separated solids are all composted on-site prior to any secondary use, so the wet weight of the separated solids is the same as the quantity of waste composted (Q_y), which is also a monitored parameter measured by direct and continuous weighing, and thus: $M_{ss,y} = Q_y * (1 - \text{moisture content}) * 1000$
Comments	-

Data / Parameter	$VS_{ss,y}$
Data unit	kg/kg
Description	Volatile solids content of the separated solids in year y on a dry matter basis
Source of data	Ex-ante values could be retrieved from the Feasibility Study Report Ex-post values are from the test report

Description of measurement methods and procedures to be applied	The volatile solids content of the separated solids is tested on a sample basis as per “Standard for sampling and surveys for CDM project activities and Programmes of Activities (Version 09.0)” ¹³ with 90/10 confidence/precision level. Testing is performed with a muffle furnace following the procedures in Annex 2 of AM0073 (Version 01) ¹⁴
Frequency of monitoring/recording	Yearly
Value applied	Ex-ante: 0.55
Monitoring equipment	Muffle furnace
QA/QC procedures to be applied	Monitoring equipment installed on-site should be calibrated by an accredited organization according to national standards
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	-

Data / Parameter	Q_y
Data unit	t/yr
Description	Quantity of waste composted in year y (wet basis)
Source of data	Ex-ante values could be retrieved from the Feasibility Study Report Ex-post values are from the measuring records.
Description of measurement methods and procedures to be applied	As per Tool 13, there are two options to determine Q_y : 1) Option 1: Procedure using a weighing device Monitor the weight of waste delivered to the composting installation using an on-site weighbridge or any other applicable and calibrated weighing device (e.g. belt-scales) 2) Option 2: Procedure without using a weighing device This procedure shall only be applied in the case that there is no weighbridge or any other applicable and calibrated weighing device available on site. Under this procedure, Q_y is

¹³ https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20210531160756223/Meth_Stano5.pdf

¹⁴ https://cdm.unfccc.int/UserManagement/FileStorage/CDMWF_AM_8CTM5MFZLTDO7SZK1A6D7AK3YPIG7S

	calculated based on the carrying capacity of each truck delivering waste to the composting installation in year y (CT_y) The project chooses Option 1 for determining Q_y. The wet weight of the solids composted are measured by direct weighing installed on-site, and the staff record the wet weight to the measuring records
Frequency of monitoring/recording	Continuously
Value applied	Ex-ante: 52,500
Monitoring equipment	Weighing device
QA/QC procedures to be applied	Monitoring equipment installed on-site should be calibrated by an accredited organization according to national standards
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	As all separated solids are composted on site prior to any secondary use, Q_y is equal to the wet weight of the separated solids taken out of the solid separation system

Data / Parameter	$EC_{PJ,compost,y}$
Data unit	MWh/yr
Description	Quantity of electricity consumed by the project electricity consumption from composting in year y (source j is composting)
Source of data	Ex-ante values could be retrieved from the Feasibility Study Report Ex-post values are from the reading records
Description of measurement methods and procedures to be applied	The quantity of electricity consumed by composting is determined based on the electricity consumption within the project boundaries ($EC_{PJ,project,y}$) measured by electricity meters installed on-site. The staff read the electricity meters and record the quantity of electricity consumption to the reading records
Frequency of monitoring/recording	Monthly
Value applied	Ex-ante: 100
Monitoring equipment	Electricity meters

QA/QC procedures to be applied	Monitoring equipment installed on-site should be calibrated by an accredited organization according to national standards
Purpose of data	Calculation of project emissions
Calculation method	$EC_{PJ,compost,y} = EC_{PJ,project,y}$
Comments	As part of the project activity, composting takes place at the project site, where the electricity consumed by the composting, the separation of solids and all related facilities within the project boundaries are measured together by two main electricity meters installed on-site, so the monitored data for composting alone would not be available. To avoid underestimating the project emissions, the electricity consumption within the project boundaries is used to determine the electricity consumption emissions caused by composting. This is intended to calculate the quantity of electricity consumption in a more accurate way by relying on continuously measurements rather than non-monitored parameters

5.3 Monitoring Plan

The monitoring plan presented in this PD assures that real, measurable, long-term GHG emission reductions can be monitored, recorded, and reported. It is a crucial procedure to identify the final VCU of the project. This monitoring plan will be implemented by the project owner during the project operation period. The details of the monitoring plan are specified as follows:

A. Data and parameters

Data and parameters to be monitored are listed below shows the corresponding parameters monitored included as following:

Monitoring Point	Parameter	Monitoring and Recording
E1	$M_{ss,y}$	The wet weight of all outputs from separation system is equal to Q_y , and its moisture content is measured through representative sampling. The moisture content is further used to calculate the dry matter content of the separated solids. The staff record moisture content to the measuring records.
E2	$VS_{ss,y}$	The volatile solids content of the separated solids is tested on a sample basis either by a muffle furnace following the procedures in Annex 2 of AM0073 (Version 01).

E1	Q_y	The wet weight of the solids composted is directly and continuously measured by weighing device installed on-site, and the staff record the wet weight to the measuring records.
E3	$EC_{PJ,compost,y}$	The quantity of electricity consumed by composting is determined based on the electricity consumption within the project boundaries ($EC_{PJ,project,y}$) measured by electricity meters installed on-site. The staff read the electricity meters and record the quantity of electricity consumption to the reading records.

B. Management Structure

The project owner organizes a specific VCS team in the project development department to be responsible for data collection, supervision, and witness to the whole process of data monitoring and recording. A VCS manager is appointed to take full responsibility for the overall monitoring of the project. The monitoring and recording of parameters listed in Section 5.2 are carried out by designated monitoring officers. In addition, the project developer appoints internal verifiers who is responsible for internal check of the measurement, collection of relevant record, and the calculation of the emission reductions. A monitoring and management manual of the project that identifies detailed duties and responsibilities of the relevant parties is developed and served as the basis of the project monitoring. Figure 5.1 shows the operation and management structure of the project.

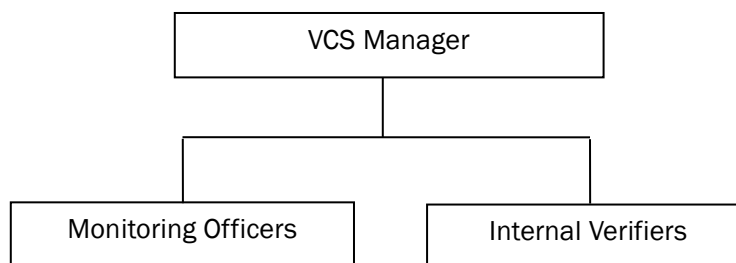


Figure 5 Operation and management structure of the project

C. Data collection

Monitoring officers are responsible for data collection. Designated teams will monitor, collect and record the monitored data regularly. The relevant production and operation data records will serve as the main data source for emission reduction calculation. All record files will be collected by a designated monitoring officer, who will prepare backup in time and archive all documents properly.

D. QA&QC

Measuring instruments should be calibrated by an accredited organization according to national standards.

E. Data file management

All monitoring data will be filed regularly, and the data files will be archived in printed hard copy or electronic disk copy. Other documents in paper e.g., maps, forms, and environment assessment reports will be preserved as well. All data collected as part of monitoring will be archived electronically and be kept at least 2 years after the end of the crediting period. The project owner will provide original records and documents if necessary.

APPENDIX